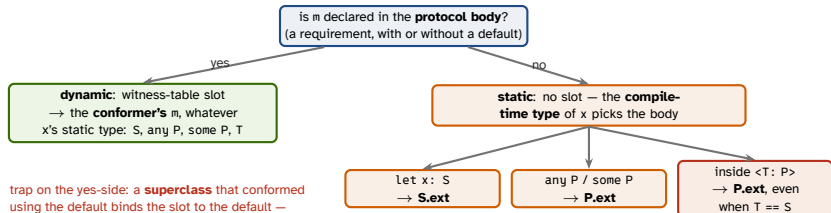


In one line: A generic signature is a set of **requirements** the compiler checks at the call site: some P in a parameter is an *unnamed* generic <T: P>, any P is a *box* for any conformer, a primary associated type (Collection<Int>) constrains either, and where adds same-type / superclass / conformance rules. Which method body runs is decided by one question: **is it a protocol requirement?** Yes → dynamic (witness table); no → the *static* type picks.

```
protocol P { func req() -> String }
extension P {
  func req() -> String { "P.req" } // default
  func ext() -> String { "P.ext" } // ext-only
}
struct S: P {
  func req() -> String { "S.req" }
  func ext() -> String { "S.ext" }
}
func gen<T: P>(_ x: T) -> String {
  x.req() + " " + x.ext()
}
gen(S()) // "S.req P.ext" (!)
let e: any P = S()
e.req(); e.ext() // "S.req", "P.ext"
S().ext() // "S.ext"
```

Which body runs for x.m()? — static vs dynamic



trap on the yes-side: a **superclass** that conformed using the default binds the slot to the default — a subclass's m is never reached via P

generics are *not* templates: T's body is type-checked once, against P only; specialisation is an optimisation and never changes which body runs. Fix: make m a requirement.

some and any — one line each

```
func draw(_ s: some Shape) // 5.7 SE-0341 = func draw<T: Shape>(_ s: T)
func draw(_ a: some Shape, _ b: some Shape) // TWO independent types
func draw(_ xs: [some Shape]) // ONE hidden type for every element
func draw(_ xs: [any Shape]) // a box per element: types may mix
func make() -> some Shape // callee picks ONE type; every return same
func make() -> any Shape // may differ per call; always boxed
func ints() -> some Collection<Int> // 5.7 SE-0346 primary associated type
let c: any Collection<Int> = Set([1]) // 5.7 SE-0353 constrained existential
func two() -> (some View, some View) // 5.7 SE-0328 structural opaque result
protocol Store<Item> { associatedtype Item // declares the primary type
let a = make(), b = make(); a == b // compiles if make() -> some Equatable
```

Opening an existential (5.7, SE-0352)

```
func area<T: Shape>(_ s: T) -> Double { s.area }
let s: any Shape = Circle(r: 1)
area(s) // OK: s is OPENED, T = its dynamic type (Circle)
drawAll(shapes) // [any Shape] to [some Shape]: error - opening is
// top-level only; 'any Shape' cannot conform to 'Shape'
func dup<T: Shape>(_ s: T) -> T; dup(s) // result erased back to any Shape
```

any P does **not** conform to P (no self-conformance) — except any Error (and some @objc protocols).

Opening is the way in.

Metatypes — Self, self, .Type, .self

```
self / Self // this instance / its type (protocol: the conformer)
func copy() -> Self // in a class: the DYNAMIC type; needs required init
Int.self // the metatype VALUE; its type is Int.Type
decode([User].self, from: d) // pass a type as an argument
type(of: view) // dynamic type: UIButton even via a UIView variable
any Shape.Type // metatype of SOME conformer: .init(), static members
(any Shape).Type // metatype of the box itself (was Shape.Protocol)
T.init() // needs an init() requirement / required init
```

Requirement shapes — what satisfies what

```
protocol Shape {
  var area: Double { get } // let, var, stored or computed all satisfy
  var name: String { get set } // needs a settable var (not let)
  static var unit: Self { get } // static: reachable via any Shape.Type
  init(size: Double) // class conformer: 'required init' (or final)
  mutating func scale(by k: Double) } // a class implements it WITHOUT mutating
  @objc protocol Legacy { @objc optional func tick() } // optional = @objc only
```

Constraints and where

```
func save<T: Codable & Identifiable>(_ x: T) // composition
func style<V: UIView & Themeable>(_ v: V) // superclass + protocol
protocol Delegate: AnyObject { } // class-only: can be weak
func merge<A: Sequence, B: Sequence>(_ a: A, _ b: B)
  where A.Element == B.Element // same-type requirement
extension Array where Element == String { } // concrete same-type
extension Collection where Element: Numeric {
  var total: Element { reduce(0, +) } } // constrained API
struct Box<T> { var v: [T]
  func sorted() -> [T] where T: Comparable { v.sorted() } } // 5.3 SE-0267
extension Box: Equatable where T: Equatable { } // conditional conformance
extension P where Self: UIViewController { } // only for VC conformers
```

Associated types · generic subscripts · ~Copyable

```
protocol Feed {
  associatedtype Item: Identifiable = Post // constraint + default
  associatedtype Page: Sequence where Page.Element == Item
  func load() async throws -> Page }
struct JSON { subscript<T: Decodable>(key: String) -> T? { nil } }
let n: Int? = json["count"] // T inferred from the context
struct Pipe<T: ~Copyable>: ~Copyable { var v: T } // 6.0 SE-0427
extension Pipe: Copyable where T: Copyable { } // copyable only if T is
func zip<each A>(_ a: repeat each A) // variadic: see the packs sheet
```

Traps

- ! [some P] is homogeneous, [any P] heterogeneous — different APIs.
- ! some P parameters can't be named or related: use <T> when two arguments must share a type.
- ! -> some P with an if returning two types: error (unless a result builder wraps it).
- ! Bare P as a type still means any P; an associated-type protocol needs any; the upcoming feature ExistentialAny requires it everywhere.
- ! On any P a member that *returns* Self/an associated type is erased to its bound (5.7, SE-0309); one that *takes* Self (==) can't be called — open the box first.

Remember

“some = the callee's one type · any = a box · requirement = dynamic, extension-only = static, even in generics.”

Related: protocols-generics (witness tables, 4 dispatch kinds) · swift-idiom-patterns (type erasure, phantom types) · parameter-packs-and-new-syntax · ownership-and-memory-layout · swift-syntax-closures-calls